Regression: establishment of a functional Relation between dependent and independent varibles, to predict one unknown value of one varible from the known value of other variable.

If x and y are two variables, then we can establish a functional Relation between x and y in 2 ways.

1. Regression of y on x (y dependent, x independent)
2. Regression of x on y. (x dependent, y independent)

Two fond Regression we use two methods.

1. By using least squares
2. Deviations taken from mean

1. By using least squares.

Regression of y on x:

$y = a_0 + a_1 x$

Normal eq: $\Sigma y = n a_0 + a_1 \Sigma x$

$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$

Regression of x on y $\quad x = a_0 + a_1 y$

Normal eqs are

$$\Sigma x = n a_0 + a_1 \Sigma y$$

$$\Sigma xy = a_0 \Sigma y + a_1 \Sigma y^2$$

Given the bi-variate data

x	10	12	13	16	17	20	25
y	10	22	24	27	29	33	37

Find the Regression lines. Find the value of x when $y = 35$.

x	y	x^2	xy	y^2
10	10	100	100	100
12	22	144	264	484
13	24	169	312	576
16	27	256	432	729
17	29	289	493	841
20	33	400	660	1089
25	37	625	925	1369
$\Sigma x = 113$	$\Sigma y = 182$	$\Sigma x^2 = 1983$	$\Sigma xy = 3186$	$\Sigma y^2 = 5188$

Regression of y on x.

$y = a_0 + a_1 x$

Normal eq: $\Sigma y = n a_0 + a_1 \Sigma x$

$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2$

$7 a_0 + 113 a_1 = 182$

~~$113 a_0 + 182 a_1 = 3186$~~

$113 a_0 + 1983 a_1 = 3186$

$a_0 = 0.82 \quad a_1 = 1.56$

$y = 0.82 + 1.56 x$

Regression eq of x on y

$x = a_0 + a_1 y$

Normal eqs:

$\Sigma x = n a_0 + a_1 \Sigma y$

$\Sigma xy = a_0 \Sigma y + a_1 \Sigma y^2$

$7 a_0 + 182 a_1 = 113$

$182 a_0 + 5188 a_1 = 3186$

$a_0 = 2.0025, \quad a_1 = 0.5439$

$x = 2.0025 + 0.5439 y$

Find x when y = 35

Regression of x on y

$x = 2.0025 + 0.5439 y$

$x = 21.0390$

Fit a linear regression equations for the following data. Calculate y at $x = 5$ (4)

x	2	4	6	8	10	12
y	1.8	1.5	1.4	1.1	1.1	0.9

x	y	x^2	y^2	xy
2	1.8	4	3.24	3.6
4	1.5	16	2.25	6
6	1.4	36	1.96	8.4
8	1.1	64	1.21	8.8
10	1.1	100	1.21	11
12	0.9	144	0.81	10.8
$\Sigma x = 42$	$\Sigma y = 7.8$	$\Sigma x^2 = 364$	$\Sigma y^2 = 10.68$	$\Sigma xy = 48.6$

Regression of y on x

$$ma_0 + a_1 \Sigma x = \Sigma y$$

$$a_0 \Sigma x + a_1 \Sigma x^2 = \Sigma xy$$

$$6a_0 + 42a_1 = 7.8$$

$$42a_0 + 364a_1 = 48.6$$

$a_0 = 1.9$, $a_1 = -0.0857$

$y = 1.9 - 0.0857x$

Regression of x on y $\quad x = a_0 + a_1 y$

$$\Sigma x = n a_0 + a_1 \Sigma y$$

$$\Sigma xy = a_0 \Sigma y + a_1 \Sigma y^2$$

$$6 a_0 + 7.8 a_1 = 42$$

$$7.8 a_0 + 10.68 a_1 = 48.6$$

$$a_0 = 21.4444, \quad a_1 = -11.1111$$

$$x = 21.4444 - 11.1111 y$$

Find y when $x = 5$

$$y = 1.9 - 0.0857 x$$

$$n = 5$$

$$y = 1.9 - 0.0857(5)$$

$$y = 1.4715$$

2nd model: mean deviations

y dependent, x independent

Regression of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where $b_{yx} = \frac{\Sigma xy}{\Sigma x^2}$

x dependent, y independent

Regression of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where $b_{xy} = \frac{\Sigma xy}{\Sigma y^2}$

Calculate the Regression eq of y on x, from the following data by taking deviations from mean.

(6)

price	10	12	13	12	16	15
Demand	40	38	43	45	37	43

$\bar{x} = 13, \quad \bar{y} = 41$

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
10	40	-3	-1	9	1	3
12	38	-1	-3	1	9	3
13	43	0	2	0	4	0
12	45	-1	4	1	16	-4
16	37	3	-4	9	16	-12
15	43	2	2	4	4	4
				$\Sigma X^2 = 24$		$\Sigma XY = -6$

$$b_{yx} = \frac{\Sigma XY}{\Sigma X^2} = \frac{-6}{24} = -\frac{1}{4} = -0.25$$

Required eq $y - \bar{y} = b_{yx}(x - \bar{x})$

~~$y - 41 = -\frac{1}{4}(x - 13)$~~

$y - 41 = -0.25(x - 13)$

$y = 41 - 0.25x + 3.25$

$y = -0.25x + 44.25$

Find a Regression line by taking deviations from mean for the following data. Also Calculate the value of x when $y = 70$ (7)

x	45	47	39	57	54	46	
y	32	29	37	26	31	30	

Regression of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x}).$$

Here b_{yx} is ~~called slope~~

$$b_{yx} = \frac{\Sigma xy}{\Sigma x^2} \quad \text{or} \quad b_{yx} = r\frac{\sigma_y}{\sigma_x}$$

similarly Regression of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

Here b_{xy} is ~~called slope~~

$$b_{xy} = \frac{\Sigma xy}{\Sigma y^2} \quad \text{or} \quad b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

Let θ be the angle between the two Regression lines, y on x and x on y Then

$$\tan\theta = \left(\frac{1-r^2}{r}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

Note: $\sigma_x^2 = \frac{\Sigma(x-\bar{x})^2}{n}$, $\sigma_y^2 = \frac{\Sigma(y-\bar{y})^2}{n}$

If $X = 37$, then $Y = 0.75\,(37) + 5.75 = 33.5$

Hence if the judge Q would have been present, he would have awarded 33.5 marks to the eight performance.

Example 4 : Find the most likely production corresponding to a rainfall 40 from the following data:

	Rain fall (X)	Production (Y)
Average	30	500 Kgs
Standard deviation	5	100 Kgs
Coefficient of correlation	0.8	

Solution : We have to calculate the value of Y when $X = 40$.

So we have to find the regression equation of Y on X.

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$\bar{x} = 30, \quad \bar{y} = 500$

$\sigma_x = 5 \qquad \sigma_y = 100$

$r = 0.8$

find production (y) corresponding to rainfall (x) 40

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 500 = r \frac{\sigma_y}{\sigma_x} (x - 30)$$

$$y - 500 = 0.8 \frac{(100)}{5} (x - 30)$$

$$y = 500 + 16(x - 30)$$

$$y = 16x + 20$$

Put $x = 40$

$$y = 16 \times 40 + 20$$

$$y = 66$$

Given $n=10$, $\sigma_x = 5.4$, $\sigma_y = 6.2$. Sum of the product of the deviations from the mean x and y is 66. Find Correlation Coefficient?

Sol: $n=10$, $\sigma_x = 5.4$, $\sigma_y = 6.2$

$$\sigma_x^2 = \frac{1}{n} \Sigma (x-\bar{x})^2$$

$$(5.4)^2 \times 10 = \Sigma x^2$$

$$\Sigma x^2 = 291.6$$

IIly $$\sigma_y^2 = \frac{1}{n} \Sigma (y-\bar{y})^2$$

$$6.2^2 \times 10 = \Sigma y^2$$

$$\Sigma y^2 = 384.4$$

Sum of the product of the deviations from the mean x and y is means

$$\Sigma (x-\bar{x})(y-\bar{y}) = 66 \Rightarrow \Sigma xy = 66$$

$$\gamma = \frac{\Sigma dx\, dy}{\sqrt{\Sigma dx^2}\sqrt{\Sigma dy^2}} \text{ or } \gamma = \frac{\Sigma xy}{\sqrt{\Sigma x^2}\sqrt{\Sigma y^2}}$$

$$= \frac{66}{\sqrt{291.6}\sqrt{384.4}} = 0.1971$$

If θ is the angle between two regression lines and S.D of y is twice the S.D of x and $\gamma = 0.25$ find $\tan\theta$.

S.D of y is twice the S.D of x

$$\sigma_y = 2\sigma_x$$

$$\tan\theta = \left(\frac{1-\gamma^2}{\gamma}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

$$= \left[\frac{1-(0.25)^2}{0.25}\right]\frac{\sigma_x\, 2\sigma_x}{\sigma_x^2 + 4\sigma_x^2}$$

$$= \frac{1-0.0625}{0.25}\;\frac{2\sigma_x^2}{5\sigma_x^2}$$

$$= 1.5$$

$$\theta = \tan^{-1}(1.5)$$

If $\sigma_x = \sigma_y = \sigma$ and the angle between the Regression lines is $\tan^{-1}\left(\frac{4}{3}\right)$. find r. (5)

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\tan\theta = \frac{4}{3}$$

$$\left(\frac{1-r^2}{r}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = \frac{4}{3}$$

$$\frac{1-r^2}{r}\frac{\sigma\sigma}{\sigma^2+\sigma^2} = \frac{4}{3}$$

$$\frac{1-r^2}{r}\cdot\frac{\sigma^2}{2\sigma^2} = \frac{4}{3}$$

$$\frac{1-r^2}{r} = \frac{8}{3}$$

$$3r^2 + 8r - 3 = 0$$

$$r = 0.333 \text{ or } r = -3$$

$$\therefore r = 0.333$$

The tangent of the angle between the two lines of regression is 0.5 and $\sigma_x = \frac{1}{4}\sigma_y$. Find the value of Correlation Coefficient?

$$\tan\theta = 0.5$$

$$\left(\frac{1-r^2}{r}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} = 0.5$$

$$\frac{1-r^2}{r}\;\frac{\frac{1}{4}\sigma_y \sigma_y}{\left(\frac{1}{4}\sigma_y\right)^2 + \sigma_y^2} = 0.5$$

$$\frac{1-r^2}{r}\;\frac{0.25\,\sigma_y^2}{1.25\,\sigma_y^2} = 0.5$$

$$\frac{1-r^2}{r} = \frac{0.5 \times 1.25}{0.25}$$

$$\frac{1-r^2}{r} = 2.5$$

$$r^2 + 2.5r - 1 = 0$$

$$r = 0.3508 \text{ or } r = -2.8508$$

$$\therefore r = 0.3508$$

y students seeking Ac

(7)

And the mean values of the variables x and y and Correlation Coefficient from the following regression equations.

$2y - x - 50 = 0$ $\quad 3y - 2x - 10 = 0$

Soln: Rewrite the eqs

$-x + 2y = 50$ ... (1)

$-2x + 3y = 10$ ... (2)

On solving we get

$\bar{x} = 130, \quad \bar{y} = 90$

(1) $\Rightarrow -x + 2y = 50$

slope $m_1 = -\frac{\text{Coef } x}{\text{Coef } y}$

$m_1 = -\frac{(-1)}{2}$

$m_1 = \frac{1}{2}$

(2) $\Rightarrow -2x + 3y = 0$

slope $m_2 = -\frac{\text{Coef } x}{\text{Coef } y}$

$m_2 = \frac{2}{3}$

$r^2 = \frac{m_1}{m_2}$

$= \frac{\frac{1}{2}}{\frac{2}{3}}$

$r^2 = \frac{3}{4}$

$r = 0.866$